Mini Project Report on

PULSE ANALYTICS PLATFORM

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BONAFIDE CERTIFICATE

Certified that this project report **Pulse Analytics Platform** is the bonafide work of **JYOTI RAJPUT** who carried out the project work under my supervision

Submitted for the project viva-voice examination held on 21 April,2026.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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TITLE OF PROJECT REPORT

Pulse Analytics Platform

ABSTRACT

The **Pulse Analytics Platform** is a full-stack web analytics system designed to track, analyze, and visualize user interactions on websites in a scalable and privacy-conscious manner. With the growing importance of data-driven decision-making, organizations require efficient tools to understand user behavior, monitor website performance, and optimize digital strategies. This project addresses these needs by providing a customizable and self-hosted analytics solution.

The platform is developed using a **Django REST Framework** backend and a **React-based dashboard** frontend. It captures key metrics such as page views, unique visitors, session duration, referrers, device types, browser usage, and geographic information. A lightweight tracking mechanism is implemented using browser storage techniques to generate unique visitor and session identifiers, enabling accurate event tracking without heavy dependencies.

The backend API efficiently collects and processes analytics data, which is then stored in a structured database. The frontend dashboard presents this data through interactive visualizations, allowing users to gain meaningful insights into traffic patterns and user engagement. The system is designed to be modular, scalable, and easy to deploy, making it suitable for both small and medium-scale applications.

The results demonstrate that the platform successfully provides real-time insights and improves the understanding of user behavior. Compared to traditional third-party analytics tools, it offers enhanced flexibility, better control over data privacy, and reduced dependency on external services.

Future enhancements may include real-time analytics using WebSockets, machine learning-based predictive insights, advanced data visualization, and cloud deployment for large-scale environments. Overall, the Pulse Analytics Platform serves as an effective and extensible solution for modern web analytics needs.

CHAPTER 1: INTRODUCTION

1.1 Identification of Client / Need / Relevant Contemporary Issue

In the modern digital ecosystem, businesses heavily depend on websites to engage users and drive growth. Understanding user behavior has become essential for improving user experience, optimizing marketing strategies, and increasing conversions.

Traditional analytics tools often come with limitations such as complexity, lack of customization, privacy concerns, and high cost. Therefore, there is a need for a customizable, lightweight, and self-hosted analytics platform.

Pulse Analytics Platform is developed to address these needs by providing real-time insights into website traffic using a full-stack approach.

1.2 Identification of Problem

Existing analytics platforms face several issues:

- Limited control over data privacy
- Complex dashboards for beginners
- Dependency on third-party tools
- Limited customization options
- High cost for advanced features

The problem is to design a scalable, customizable, and privacy-focused web analytics system.

1.3 Identification of Tasks

The following tasks were identified:

- Design backend API for collecting analytics data
- Develop frontend dashboard for visualization
- Implement tracking mechanism for user activity
- Store and process large volumes of data
- Display meaningful insights using charts and metrics

1.4 Timeline

Phase	Duration
Requirement Analysis	1 Week
Design	1 Week
Development (Backend + Frontend)	2 Weeks
Testing	1 Week
Deployment	1 Week

1.5 Organization of the Report

- Chapter 1: Introduction
- Chapter 2: Literature Review
- Chapter 3: Design Flow
- Chapter 4: Results and Validation
- Chapter 5: Conclusion and Future Work

CHAPTER 2: LITERATURE REVIEW / BACKGROUND STUDY

2.1 Timeline of the Reported Problem

- Early 2000s: Basic web logs used for analytics
- Mid 2000s: Emergence of advanced analytics platforms
- Present: Focus on privacy, real-time analytics, and customization

2.2 Existing Solutions

Popular analytics tools include:

- Google Analytics
- Matomo
- Adobe Analytics

Limitations:

- Data privacy concerns
- Complex interfaces
- Limited customization
- Paid features for advanced analytics

2.3 Bibliometric Analysis

Studies show increasing demand for:

- Privacy-focused analytics
- Real-time data visualization
- Self-hosted analytics platforms

Research trends highlight the importance of data-driven decision-making in organizations.

2.4 Review Summary

Existing systems provide powerful features but lack flexibility, transparency, and simplicity. There is a gap for a developer-friendly analytics system.

2.5 Problem Definition

To develop a full-stack analytics platform that:

- Tracks website traffic efficiently
- Provides real-time insights
- Ensures data privacy
- Is customizable and scalable

2.6 Goals / Objectives

- Build a Django REST API backend
- Create a React-based dashboard
- Track user sessions and page views
- Analyze device, browser, and geographic data
- Visualize analytics using interactive UI

CHAPTER 3: DESIGN FLOW / PROCESS

3.1 Evaluation & Selection of Specifications / Features

Selected features include:

- Page view tracking
- Unique visitor identification
- Session tracking
- Device & browser analytics
- Geographic data visualization

3.2 Design Constraints

- Limited system resources
- Need for real-time performance
- Data accuracy requirements
- Security and privacy concerns

3.3 Analysis of Features and Finalization Subject to Constraints

Features were selected based on:

- Performance efficiency
- Ease of implementation
- Scalability
- User experience

3.4 Design Flow

Workflow:

1. User visits website
2. Tracking script collects data
3. Data sent to backend API
4. Backend stores data in database
5. Dashboard fetches data via API
6. Data visualized in frontend

3.5 Design Selection

Architecture:

- **Frontend:** React (Vite)
- **Backend:** Django REST Framework
- **Database:** SQLite / PostgreSQL

Components:

- Tracking Script
- API Layer
- Database
- Dashboard UI

3.6 Implementation Plan / Methodology

Backend:

- REST API endpoints for data collection and dashboard
- Data models for analytics events

Frontend:

- Interactive dashboard using React
- Charts and graphs for visualization

Tracking:

- LocalStorage for visitor ID
- SessionStorage for session ID
- Event-based tracking

CHAPTER 4: RESULTS ANALYSIS AND VALIDATION

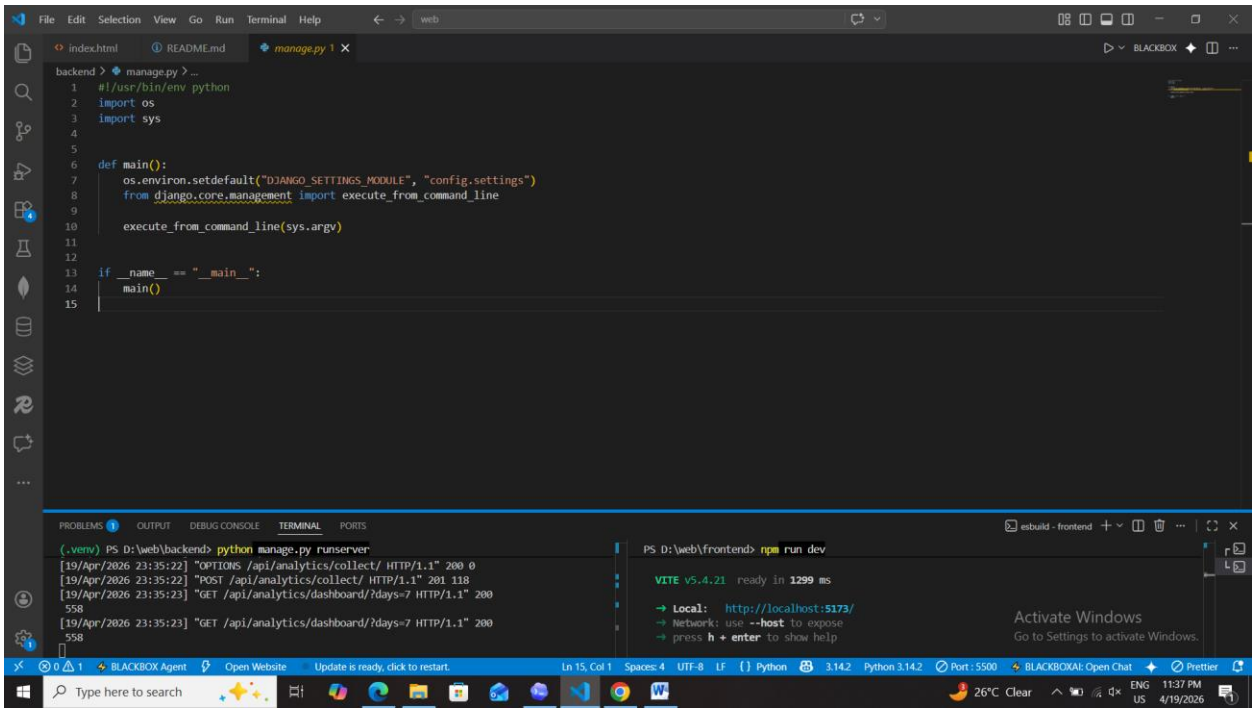
4.1 Implementation of Solution

The system was successfully implemented with:

Backend Features:

- Event collection API
- Dashboard analytics API
- Data storage and processing

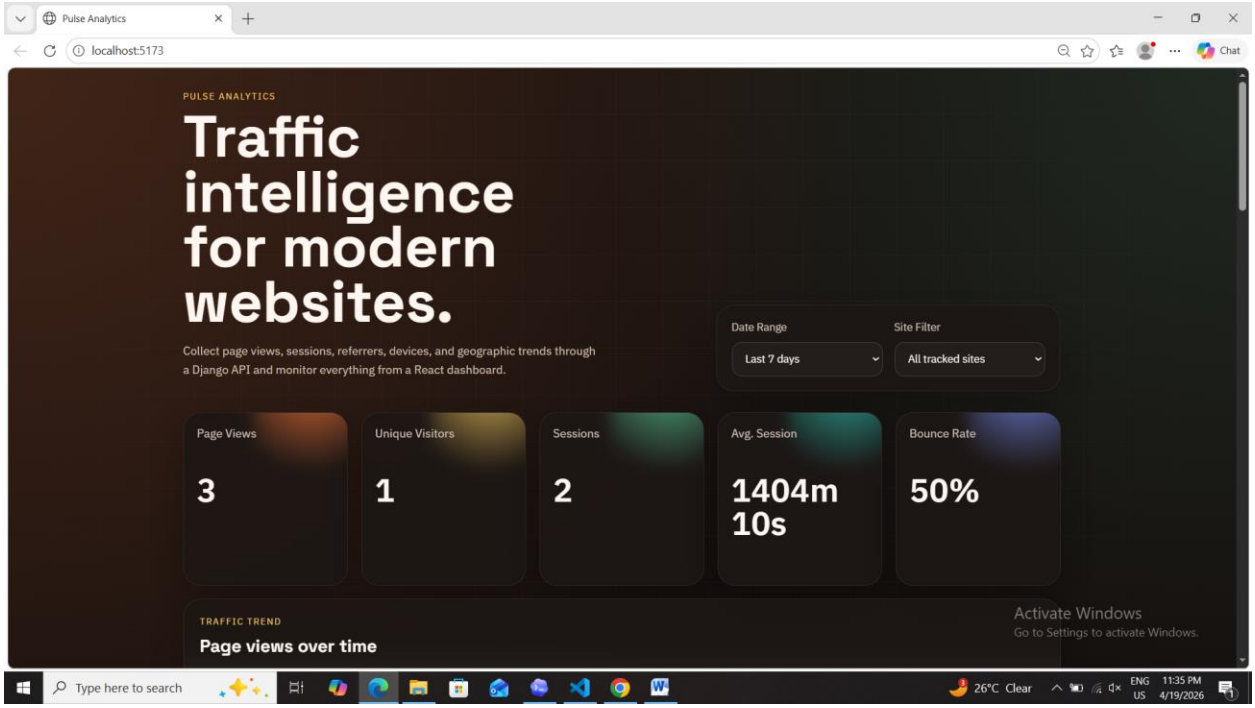
Screenshot:

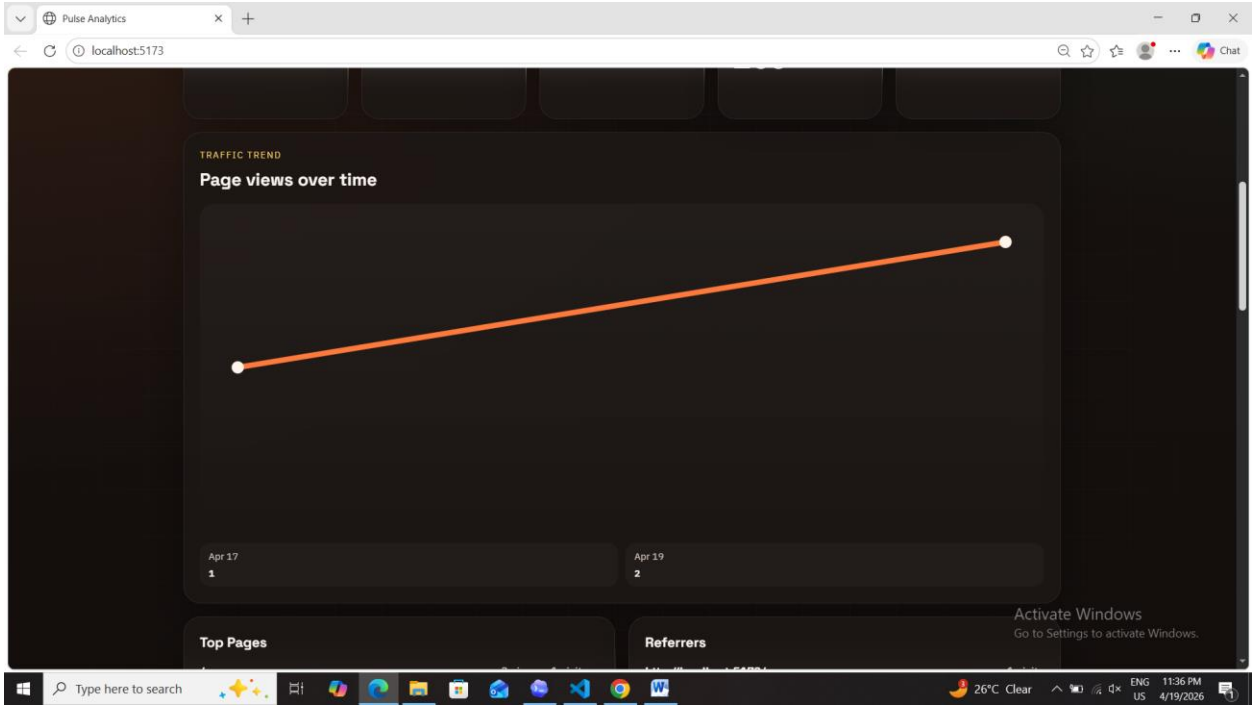


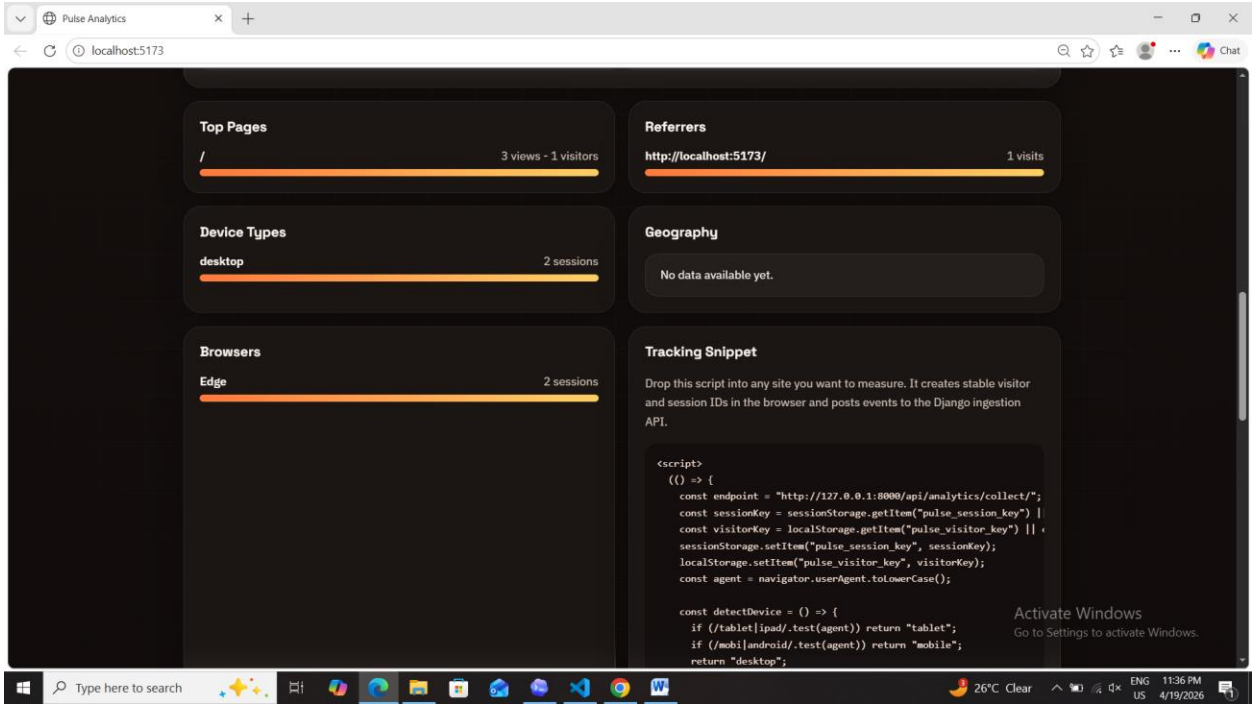
Frontend Features:

- Real-time dashboard
- Visualization of:
  - Page views
  - Unique visitors
  - Session duration
  - Device types
  - Browser usage
  - Geographic data

Screenshot:


Pulse Analytics
localhost:5173
PULSE ANALYTICS
Traffic intelligence for modern websites.
Collect page views, sessions, referrers, devices, and geographic trends through a Django API and monitor everything from a React dashboard.
Date Range
Last 7 days
Site Filter
All tracked sites
Page Views
3
Unique Visitors
1
Sessions
2
Avg. Session
1404m 10s
Bounce Rate
50%
TRAFFIC TREND
Page views over time
Activate Windows
Go to Settings to activate Windows.
Type here to search
26°C Clear
ENG US
11:35 PM
4/19/2026


Pulse Analytics
localhost:5173
TRAFFIC TREND
Page views over time
Apr 17
1
Apr 19
2
Top Pages
Referrers
Activate Windows
Go to Settings to activate Windows.
Type here to search
26°C Clear
ENG US
11:36 PM
4/19/2026



Sample API Endpoint

```
POST /api/analytics/collect/
GET /api/analytics/dashboard/?days=7
```

Sample Output

- Total Page Views: Displayed in charts
- Active Users: Count of visitors
- Session Duration: Average time spent
- Top Pages: Most visited URLs
- Device Breakdown: Desktop vs Mobile

Validation

- Tested API with sample payload
- Verified data storage in database
- Validated dashboard visualization
- Checked performance under multiple requests

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion

The Pulse Analytics Platform successfully demonstrates a scalable and efficient web analytics system. It provides valuable insights into user behavior while maintaining flexibility and privacy.

The system reduces dependency on third-party tools and offers a customizable solution for developers and organizations.

5.2 Future Work

- Add authentication and role-based access
- Implement real-time analytics using WebSockets
- Integrate machine learning for predictive analytics
- Use PostgreSQL for large-scale deployment
- Enhance UI with advanced charts
- Add anomaly detection for unusual traffic

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APPENDIX

1 Plagiarism Report

2 Design Checklist

Design Checklist

- Define system requirements (analytics metrics, users, features)
- Choose tech stack (Django REST + React + Database)

- Design system architecture (Frontend ↔ Backend ↔ Database)
- Create database schema (events, sessions, visitors)
- Design API endpoints (data collection & dashboard)
- Plan tracking mechanism (visitor ID, session ID)
- Design UI/UX for dashboard (charts, reports)
- Ensure scalability (modular structure, optimized queries)
- Implement security (authentication, data protection)
- Test system (API, UI, performance)

GITHUB LINK: https://github.com/jrajput1718/Pulse-Analytics-Platform